

机器学习范式

Machine Learning Paradigm

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声明

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学习

Learning

通过研习、实践或他人传授而掌握知识
Get knowledge of by study, experience, or be taught

通过信息获取或
观察感知而知晓
情况
Become aware by
information or
from observation

牢记于心
Commit to
memory

有所洞见或施用于外部
Be informed of or
produce instruction

主体通过学习，改变自身行为，从而在未来表现得更出色。
Subjects by learning change their behavior in a way that makes
them perform better in the future

机器学习

Machine Learning

- 广义而言，学习机器是指任何行为会受过往经验影响的装置。（尼尔森，1965）

A learning machine, broadly defined is any device whose actions are influenced by past experiences. (Nilsson 1965)

- 当一个系统在重复执行相同任务，或执行取自同一任务集合的另一项任务时，能够表现得更出色，其产生的任何此类变化均称为学习。（西蒙，1983）

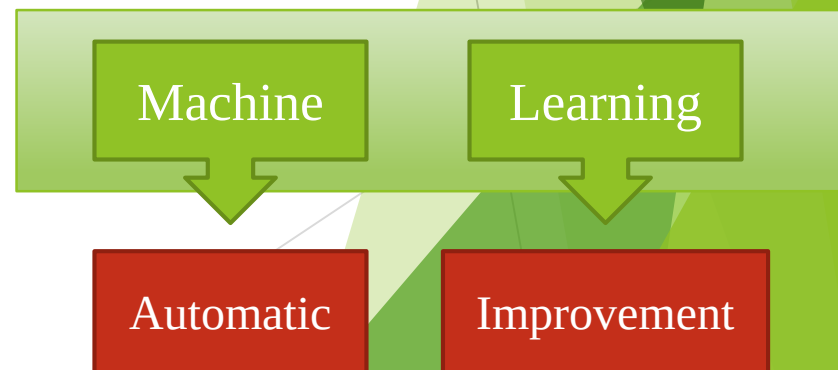
Any change in a system that allows it to perform better the second time on repetition of the same task or on another task drawn from the same population. (Simon 1983)

- 通过专业经验对行为倾向产生的修正作用。（韦伯斯特，1984）

Modification of a behavioral tendency by expertise. (Webster 1984)

- 由信息处理活动所带来的信息处理能力的提升。（谷本，1990）

An improvement in information processing ability that results from information processing activity. (Tanimoto 1990)



机器学习分类

Categories of Machine Learning (ML)

监督学习

Supervised Learning

半监督学习

Semi-supervised
Learning

无监督学习

Unsupervised Learning

强化学习

Reinforcement Learning

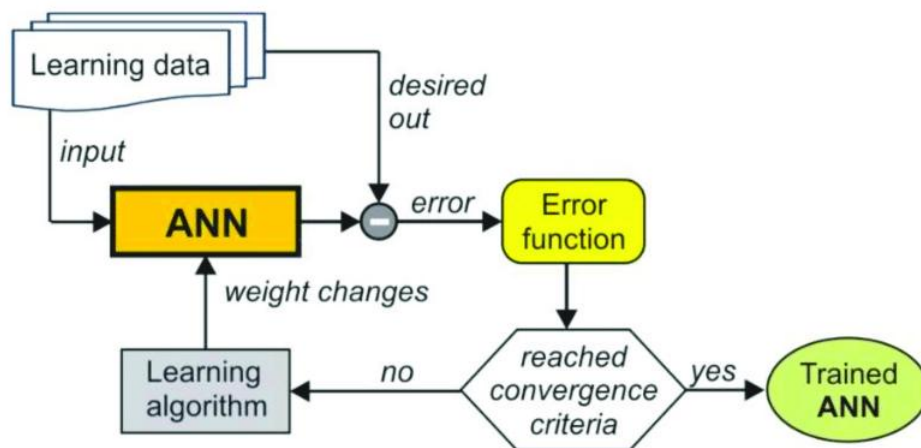
机器学习分类

Categories of Machine Learning (ML)

► 监督学习 (Supervised Learning)

- 监督学习是通过提供输入数据和标签引导计算机学习来完成的。例如，可以向机器提供带有相应标签（例如小猫）的图片，引导计算机学习小猫的内在特征，以便进行后续的泛化。之后，如果我们提供了另一只没有标签的小猫的图片，则机器将能够预测该图像就是小猫的图像。监督学习的主要任务包括分类和回归。

Supervised Learning is a learning paradigm based on example input data with corresponding ground-truth labels to train the models into optimal states for inference. Tasks for supervised learning includes classification and regression.



机器学习分类

Categories of Machine Learning (ML)

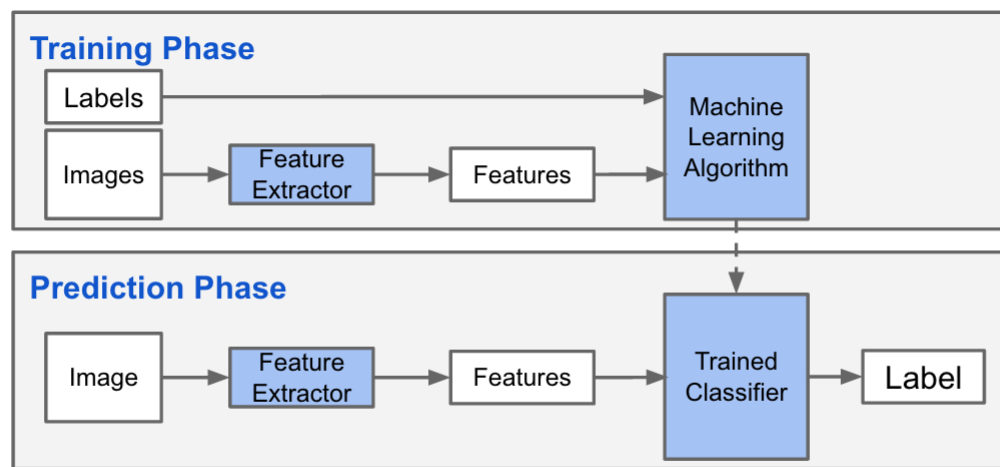
► 监督学习 (Supervised Learning)

- 监督学习通常包括两个阶段：训练和测试阶段。

The process of applying SL consists of two stages: training and testing.

- 基于监督学习的例子包括支持向量机，神经网络等

Algorithms and techniques used for supervised learning include support vector machine (SVM), neural networks (NN), etc.



Machine Learning Phases

机器学习分类

Categories of Machine Learning (ML)

► 半监督学习 (Semi-supervised Learning)

是机器学习领域中介于监督学习与无监督学习之间的一种学习范式。它利用少量有标签数据和大量无标签数据联合训练模型。

A paradigm in machine learning that lies between supervised learning and unsupervised learning. It leverages a small amount of labeled data and a large amount of unlabeled data to train models.

► 半监督学习假设无标签数据中蕴含着数据分布的内在结构信息，这些信息能够辅助模型捕捉更具泛化性的模式，从而提升模型性能——尤其适用于有标签数据稀缺或标注成本高昂的场景（如医学影像标注、自然语言处理语料库构建）。

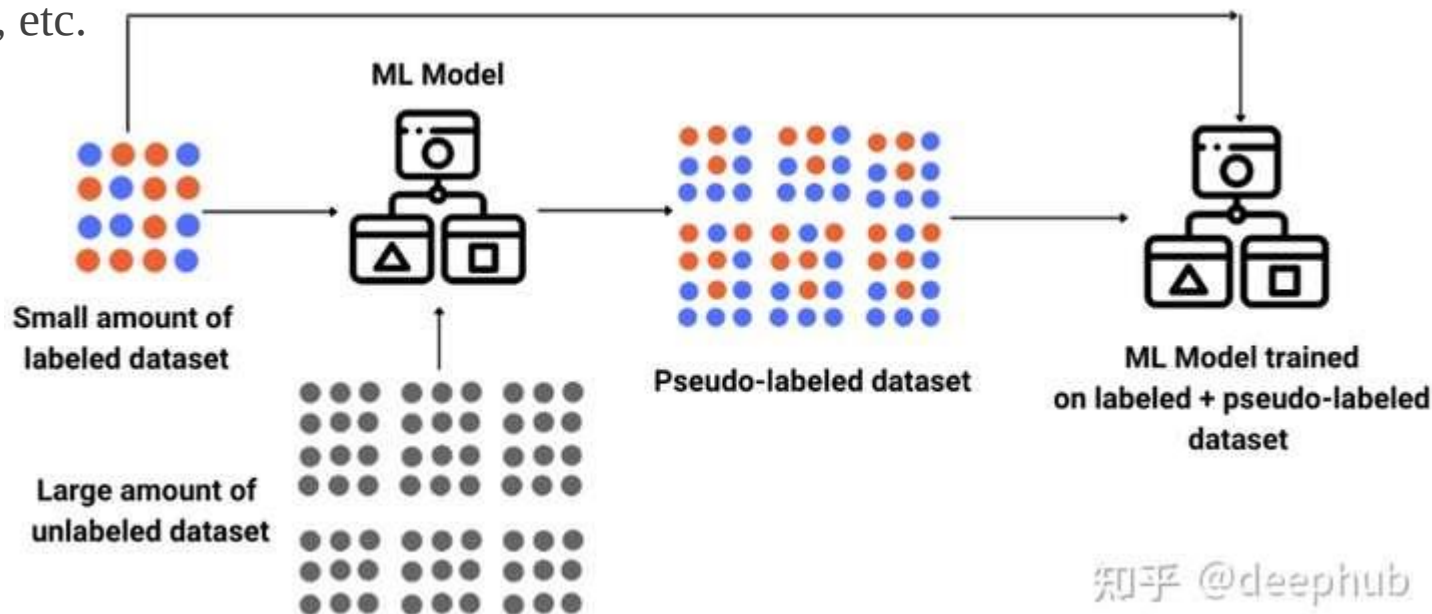
The core assumption is that unlabeled data contains inherent structural information about the data distribution, which can help the model capture more generalized patterns and improve performance—especially when labeled data is scarce or expensive to obtain (e.g., medical image annotation, natural language processing corpus construction).

机器学习分类

Categories of Machine Learning (ML)

- 半监督学习依赖的理论基础包括聚类假设（同一聚类中的数据样本大概率属于同一类别）与流形假设（数据分布于嵌入高维空间的低维流形上）等。

Common theoretical foundations underpinning semi-supervised learning include the cluster assumption (data points in the same cluster tend to have the same label) and the manifold assumption (data lies on a low-dimensional manifold embedded in a high-dimensional space), etc.



机器学习分类

Categories of Machine Learning (ML)

► 非监督学习 (Unsupervised Learning)

- 聚类分析主要用于根据数据的内在属性分组；降维主要用于减小特征的维度，创建更简明的机器学习模型。这两种方法均可用于如数据的可视化。注意非监督学习得到的结果并不准确或唯一，主要，因此要求待解决的问题具有一定的容错性。

Clustering is mainly used for grouping or categorization, and dimension reduction is to merge correlated features to create more concise machine learning models. Both methods can be used for data visualization. Notably, Unsupervised Learning usually leads to results which is not unique, therefore, it tends to require that the problems to be solved can tolerant the imprecision;

- 基于非监督学习的例子包括K均值聚类，主成分分析，自组织映射等

Algorithms and techniques used for unsupervised learning include K-means clustering, principal component analysis (PCA), self-organizing maps (SOM), etc.

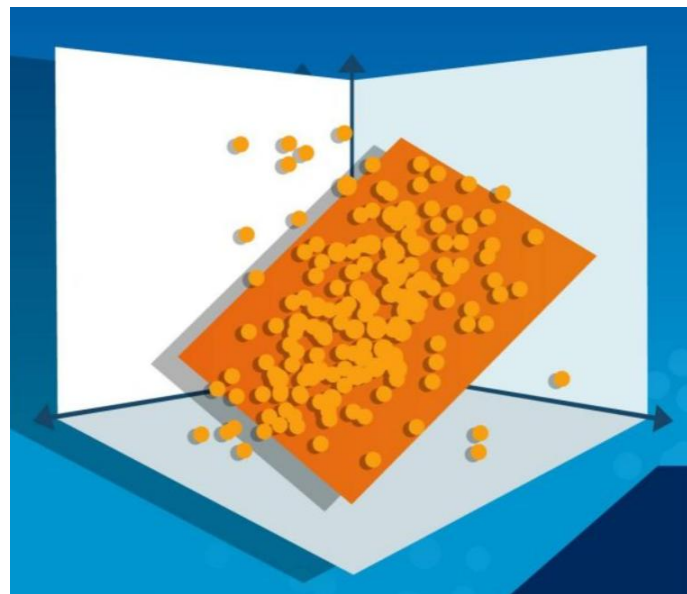
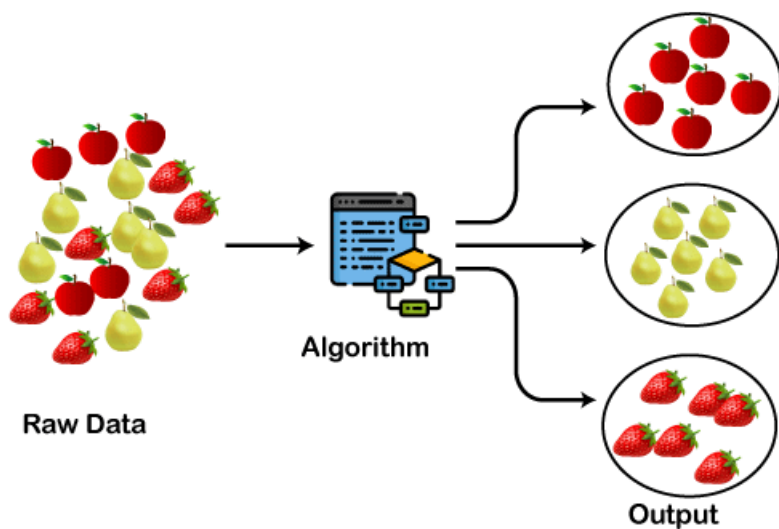
机器学习分类

Categories of Machine Learning (ML)

► 非监督学习 (Unsupervised Learning)

- 无监督学习是从不包含标签的数据集中得出推论的算法，主要包括聚类分析和降维等方法。

Unsupervised learning (UL) is a type of learning that learns patterns from untagged data by building an appropriate internal representation of the data. Methods belonging to UL include clustering and dimension-reduction.



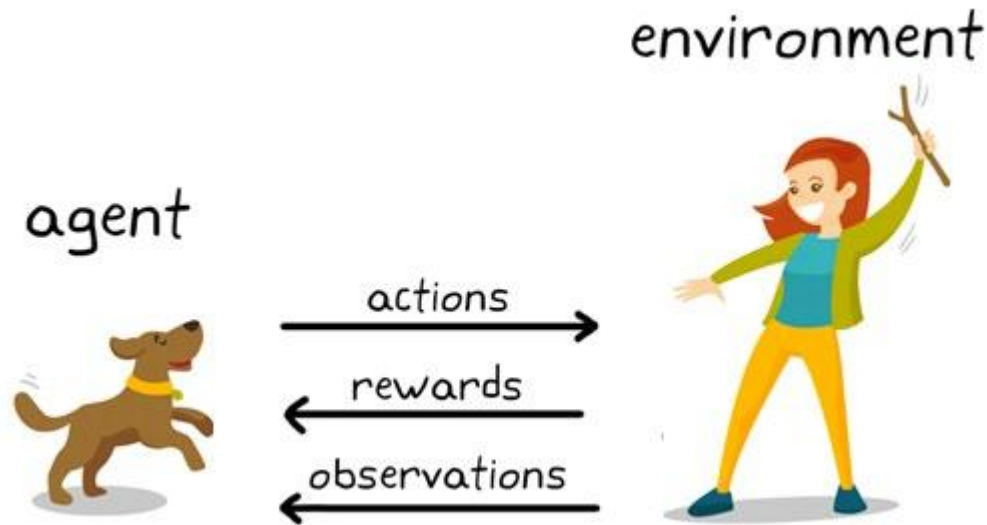
机器学习分类

Categories of Machine Learning (ML)

► 强化学习 (Reinforcement Learning)

- 强化学习是智能体通过与环境进行交互，即执行某些操作并观察从这些操作中获得的奖励来学习如何响应环境中的特定状态。

Reinforcement learning (RL) concerns with the approaches that how intelligent agents effectively interact with the environments by taking actions.



机器学习分类

Categories of Machine Learning (ML)

► 强化学习 (Reinforcement Learning)

- 强化学习主要通过试错方式，学习从环境状态到动作的映射，以使动作从环境中获得的累积奖励最大化。

By a trial-and-error way, the main goal of reinforcement learning is to learn the optimal mapping between states and the corresponding actions, in order to maximize the notion of cumulative reward in the long run.

- 强化学习主要包括算法主要包括Q学习算法，动作器-评判器算法，近端策略优化算法等等。

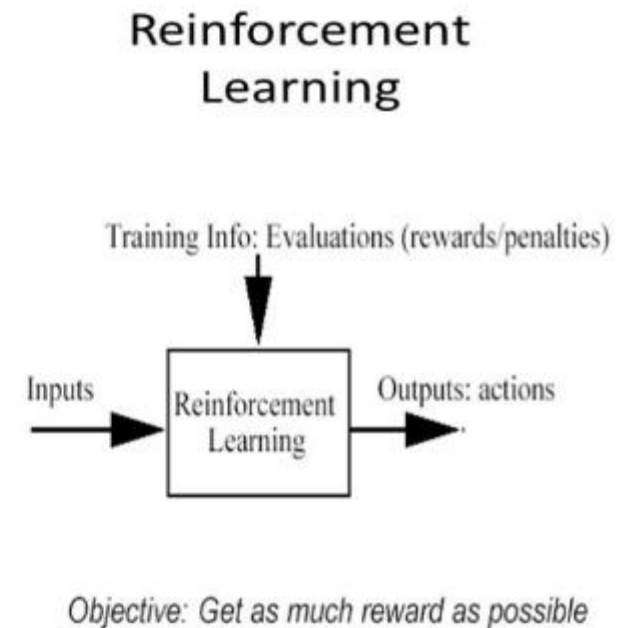
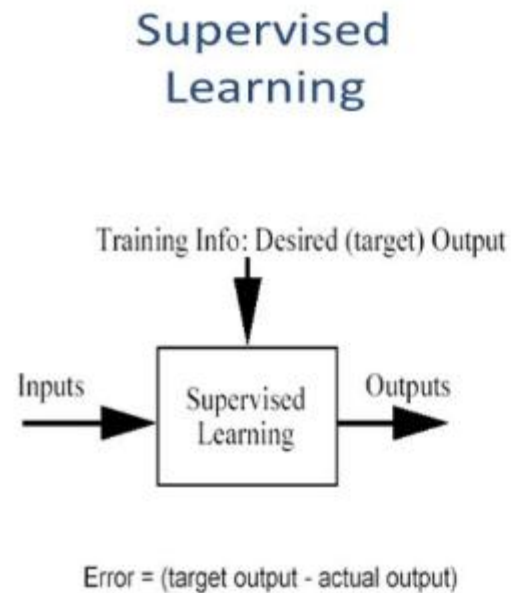
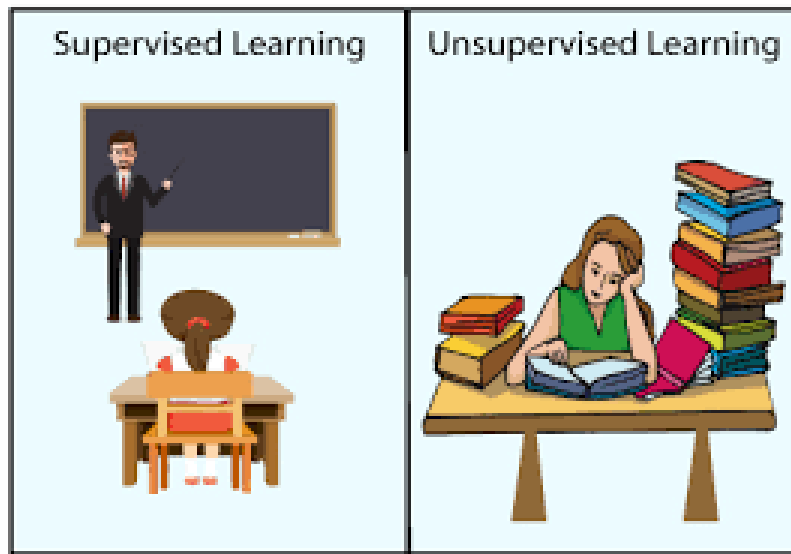
Methods of reinforcement learning includes Q-Learning, Actor-Critic algorithm and Proximal Policy Optimization method, etc.

机器学习分类

Categories of Machine Learning (ML)

► 三种学习方式的比较

Comparison of these three learning paradigms



机器学习要素

Elements of Machine Learning (ML)

► 解决机器学习问题的组成要素：

Elements of machine learning problems:

❖ 数据（data）；

数据即模型输入，数据的特征取决于问题领域（Domain）；

Data are the inputs to the models, might contain ground-truth labels or annotations. The features of the data are domain dependent.

❖ 模型（model）

模型即解决问题所采用的算法；

Model are the adopted algorithms to solve the problems.

❖ 目标或损失函数（Loss function）；

目标或损失函数依机器学习的范式而定，如果是监督学习，则一般有显示的（Explicit）损失函数，若是非监督学习，损失内蕴于（Intrinsic）数据或样本本身

机器学习要素

Elements of Machine Learning (ML)

Loss function usually depends on the learning paradigm. For supervised learning, there exists explicit loss function by comparing the model outputs with ground-truth information. For unsupervised learning, the loss is intrinsic to data structures or representations.

- ❖ 学习方法 (Learning algorithm) ；

学习方法即使为使模型达到目标采取行动时所依照的法则

Learning method is the rules to take for the model to reach optimal performance during learning.